



USN

|  |  |  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|--|--|

## NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

### Fifth Semester B.E. (E&E) Supplementary Degree Examinations

Supplementary Examinations - December 2025

#### 21EE503 – GENERATION, TRANSMISSION AND DISTRIBUTION

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

| Unit – I  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks | BT* | CO* | PO* |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|-----|
| 1.        | a) Discuss the choice of site for hydroelectric power plant                                                                                                                                                                                                                                                                                                                                                                                                                                             | 6     | L*2 | 1   | 1,2 |
|           | b) A generating station has a connected load of 43MW and a maximum demand of 20 MW; the units generated being $61.5 \times 10^6$ per annum. Calculate (i) the demand factor and (ii) load factor.                                                                                                                                                                                                                                                                                                       | 8     | L3  | 2   | 1,2 |
|           | c) Define nuclear fission and nuclear chain reaction.                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 6     | L1  | 1   | 1,2 |
| 2.        | a) With neat sketch describe the schematic arrangement of thermal power plant.                                                                                                                                                                                                                                                                                                                                                                                                                          | 8     | L2  | 1   | 1,2 |
|           | b) Define Tariff and explain any two types of tariffs.                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 6     | L2  | 2   | 1,2 |
|           | c) A diesel station supplies the following loads to various consumers:<br>Industrial consumer = 1500 kW; Commercial establishment = 750 kW<br>Domestic power = 100 kW; Domestic light = 450 kW<br>If the maximum demand on the station is 2500 kW and the number of kWh generated per year is $45 \times 10^5$ , determine (i) the diversity factor and (ii) annual load factor.                                                                                                                        | 6     | L3  | 2   | 1,2 |
| 3.        | a) Explain with neat schematic, the arrangement and operation of gas turbine plants.                                                                                                                                                                                                                                                                                                                                                                                                                    | 8     | L2  | 1   | 1,2 |
|           | b) A consumer has an average demand of 400 kW at a p.f. of 0.8 lagging and annual load factor of 50%. The tariff is Rs 50 per kVA of maximum demand per annum plus 5 paise per kWh. If the power factor is improved to 0.95 lagging by installing phase advancing equipment, calculate:<br>(i) the capacity of the phase advancing equipment<br>(ii) the annual saving effected<br>The phase advancing equipment costs Rs 100 per kVAR and the annual interest and depreciation together amount to 10%. | 8     | L4  | 2   | 1,2 |
|           | c) Draw the typical single line diagram of power system                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 4     | L2  | 2   | 1,2 |
| Unit – II |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |     |     |     |
| 4.        | a) Define sag. Explain the effect of wind and ice loading on the sag.                                                                                                                                                                                                                                                                                                                                                                                                                                   | 6     | L2  | 3   | 1,2 |
|           | b) A 3-phase transmission line is being supported by three disc insulators. The potentials across top unit (i.e., near to the tower) and middle unit are 8 kV and 11 kV respectively. Calculate (i) the ratio of capacitance between pin and earth to the self-capacitance of each unit (ii) the line voltage and (iii) string efficiency                                                                                                                                                               | 8     | L4  | 3   | 1,2 |
|           | c) Derive the expression for sending end voltage and current of a medium (T model) transmission line.                                                                                                                                                                                                                                                                                                                                                                                                   | 6     | L3  | 4   | 1,2 |
| 5.        | a) Derive the expression for sag when conductors are suspended on supports at equal levels.                                                                                                                                                                                                                                                                                                                                                                                                             | 6     | L3  | 3   | 1,2 |

P.T.O.

21EE503

## Supplementary Examinations - December 2025

- b) The three conductors of a 3-phase line are arranged at the corners of a triangle of sides 2 m, 2.5 m and 4.5 m. Calculate the inductance per km of the line when the conductors are regularly transposed. The diameter of each conductor is 1.24 cm. 8 L4 3 1,2
- c) Discuss the advantages and disadvantages of corona and also the methods to reduce corona. 6 L2 4 1,2
6. a) Derive the expression for inductance of a single-phase two-wire line 8 L3 3 1,2
- b) Enumerate the advantages of bundled conductors. 4 L1 3 1,2
- c) A balanced 3-phase load of 30 MW is supplied at 132 kV, 50 Hz and 0.85 p.f. lagging by means of a transmission line. The series impedance of a single conductor is  $(20 + j5)$  ohms and the total phase-neutral admittance is  $315 \times 10^{-6}$  Siemen. Using nominal T method, determine:
- (i) the A, B, C and D constants of the line (ii) sending end voltage 8 L4 4 1,2
- Unit – III**
7. a) Discuss the types and materials used for underground cables. Also discuss the properties of the insulating materials used. 8 L2 5 1,2
- b) Derive the expression for insulation resistance of single core cable. 6 L3 5 1,2
- c) State the advantages and disadvantages of HVDC transmission. 6 L1 5 1,2
8. a) Derive the expression for maximum and minimum dielectric stress in a single core cable. 8 L3 5 1,2
- b) The maximum and minimum stresses in the dielectric of a single core cable are 40 kV/cm (r.m.s.) and 10 kV/cm (r.m.s.) respectively. If the conductor diameter is 2 cm, find: (i) thickness of insulation (ii) operating voltage. 6 L4 5 1,2
- c) Differentiate radial and ring main system of ac distribution. 6 L2 5 1,2

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

\*\*\*\*\*